

CSE421 Project

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1. Introduction

This project focuses on designing a network infrastructure for FIFA World Cup hotels across six countries: France, Spain, Germany, Argentina, Brazil, and Portugal. Each hotel is represented by a router and must support communication between staff, servers, and services such as email, web hosting, and DNS.

The goal is to build a scalable and efficient network using VLSM, dynamic and static routing, and proper server configuration in Cisco Packet Tracer.

2. Network Requirements Overview

Countries and Hosts

- France: 200 Hosts
- Spain: 100 Hosts
- Germany: 150 Hosts
- Argentina: 200 Hosts
- Brazil: 250 Hosts
- Portugal: 300 Hosts

Key Design Constraints

- Each hotel is represented by a router.
- Brazil connects only to Argentina.
- Argentina, Spain, and France are interconnected via a switch.
- European loop: France → Portugal → Spain → Germany → France.
- Email server in each hotel.
- Web servers only in Argentina and France.
- Central DNS server in Argentina.
- DHCP:
 - France serves all except Argentina & Brazil.
 - Argentina & Brazil have their own DHCP servers.

3. Network Addressing (VLSM)

Base Network

Student ID: 22101321

Last 4 digits: 1321 → Split: 13 and 21

Base Network: 13.21.0.0/16

Subnet Allocation (VLSM)

Country	Hosts Needed	Subnet	Prefix
Portugal	300	13.21.0.0	/23
Brazil	250	13.21.2.0	/24
France	200	13.21.3.0	/24
Argentina	200	13.21.4.0	/24
Germany	150	13.21.5.0	/24
Spain	100	13.21.6.0	/25

4. Topology Design

- Routers represent each country.
- A switch connects Argentina, Spain, and France.
- A loop exists among European countries.
- Brazil connects only to Argentina.

5. Server Configuration

Email Servers

Each country has its own mail server:

- mail.france.com
- mail.spain.com
- mail.germany.com
- mail.argentina.com
- mail.brazil.com
- mail.portugal.com

Web Servers

- Argentina: www.argentina.fifa.com (13.21.4.3)
- France: www.france.fifa.com (13.21.3.3)

6. Routing Configuration

Dynamic Routing (RIP)

Used between:

- France
- Spain
- Germany

Static Routing

Used in:

- Argentina
- Portugal

Default Route

- Brazil routes all traffic via Argentina

Recursive Routing

- Portugal uses alternate path if France link fails (via Spain)

7. IP Address Table (Sample)

Device	IP Address
France Router	13.21.3.1
Spain Router	13.21.6.1
Germany Router	13.21.5.1

Argentina Router 13.21.4.1

Brazil Router 13.21.2.1

Portugal Router 13.21.0.1

8. Work Distribution

Member Name	Student ID	Task
Mohsen	22201480,	VLSM & Addressing, Server Configuration
Kadir, Rowdro,Mohsen	21301733, 24101226, 22201480	Topology Design
Labonno, Kadir, Rowdro	22101321, 21301733, 24101226	Routing Configuration
-	-	Testing & Documentation

9. Conclusion

This project demonstrates a complete enterprise-level network design using Cisco Packet Tracer. By implementing VLSM, dynamic and static routing, and centralized services like DNS and DHCP, the network ensures efficient communication among all FIFA hotel locations.

The design also provides redundancy through loop topology and recursive routing, ensuring reliability even in case of link failures.

10. Configuration Commands

France Router

```
enable
configure terminal
hostname FRANCE
```

```
interface g0/0
ip address 13.21.3.1 255.255.255.0
no shutdown

interface g0/1
ip address 13.21.10.1 255.255.255.252
no shutdown

interface g0/2
ip address 13.21.10.5 255.255.255.252
no shutdown

router rip
version 2
no auto-summary
network 13.21.3.0
network 13.21.10.0

end
write memory
```

Spain Router

```
enable
configure terminal
hostname SPAIN

interface g0/0
ip address 13.21.6.1 255.255.255.128
no shutdown

interface g0/1
ip address 13.21.10.2 255.255.255.252
no shutdown

interface g0/2
ip address 13.21.10.9 255.255.255.252
no shutdown

router rip
version 2
no auto-summary
network 13.21.6.0
network 13.21.10.0
```

```
end
write memory
```

Germany Router

```
enable
configure terminal
hostname GERMANY
```

```
interface g0/0
ip address 13.21.5.1 255.255.255.0
no shutdown
```

```
interface g0/1
ip address 13.21.10.6 255.255.255.252
no shutdown
```

```
interface g0/2
ip address 13.21.10.10 255.255.255.252
no shutdown
```

```
router rip
version 2
no auto-summary
network 13.21.5.0
network 13.21.10.0
```

```
end
write memory
```

Argentina Router

```
enable
configure terminal
hostname ARGENTINA
```

```
interface g0/0
ip address 13.21.4.1 255.255.255.0
no shutdown
```

```
interface g0/1
ip address 13.21.10.21 255.255.255.252
```



```
no shutdown
```

```
ip route 13.21.3.0 255.255.255.0 13.21.10.22  
ip route 13.21.5.0 255.255.255.0 13.21.10.22  
ip route 13.21.6.0 255.255.255.128 13.21.10.22
```

```
end  
write memory
```

Brazil Router

```
enable  
configure terminal  
hostname BRAZIL
```

```
interface g0/0  
ip address 13.21.2.1 255.255.255.0  
no shutdown
```

```
interface g0/1  
ip address 13.21.10.22 255.255.255.252  
no shutdown
```

```
ip route 0.0.0.0 0.0.0.0 13.21.10.21
```

```
end  
write memory
```

Portugal Router

```
enable  
configure terminal  
hostname PORTUGAL
```

```
interface g0/0  
ip address 13.21.0.1 255.255.254.0  
no shutdown
```

```
interface g0/1  
ip address 13.21.10.13 255.255.255.252  
no shutdown
```

```
interface g0/2
```

```
ip address 13.21.10.17 255.255.255.252
no shutdown

ip route 13.21.3.0 255.255.255.0 13.21.10.14
ip route 13.21.3.0 255.255.255.0 13.21.10.18

end
write memory
```